Ryota Komatsu

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Interests

Large speech-language models, reinforcement learning.

Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology)
Ph.D. Student, Engineering — Apr 2025 – Mar 2028 (expected)
Specialization: Spoken language processing
Advisor: Prof. Takahiro Shinozaki

M.Eng. — Mar 2023
Specialization: Spoken language processing
Advisor: Prof. Takahiro Shinozaki

B.Eng. — Mar 2021
Specialization: Mathematical optimization
Advisor: Prof. Isao Yamada

Research Experience

- Identified speaker dominance as a key limitation of a previous syllabic tokenization method
- Proposed a linguistically faithful syllabic tokenization method for speech language models
- Built a data curation pipeline for real and synthetic speech (filtering for audio quality, disfluency)
- Finetuned Qwen2.5-7B on 130k hours of English speech across 32 NVIDIA H100 GPUs
- Outperformed Z.ai GLM-4-Voice by 5% in semantic understanding while using 42× less training compute

Publications

Speech language models

- **Speaker-Disentangled Chunk-Wise Regression for Syllabic Tokenization**
  R. Komatsu, K. Kawakita, T. Okamoto, and T. Shinozaki
  IEEE Open Journal of Signal Processing (OJSP), 2026
- **Self-Supervised Syllable Discovery Based on Speaker-Disentangled HuBERT**
  R. Komatsu and T. Shinozaki
  IEEE Spoken Language Technology Workshop (SLT), 2024

Spoken dialogue systems based on reinforcement learning

- **Automatic Spoken Language Acquisition Based on Observation and Dialogue**
  R. Komatsu, S. Gao, W. Hou, M. Zhang, T. Tanaka, K. Toyoda, Y. Kimura, K. Hino, Y. Iwamoto, K. Mori, T. Okamoto, and T. Shinozaki
  IEEE Journal of Selected Topics in Signal Processing (JSTSP), 2022
- **Continuous Action Space-Based Spoken Language Acquisition Agent Using Residual Sentence Embedding and Transformer Decoder**
  R. Komatsu, Y. Kimura, T. Okamoto, and T. Shinozaki
  IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023

Signal processing

- **A Graph Regularized RPCA by Generalized Moreau Enhanced Model**
  R. Komatsu, M. Yamagishi, and I. Yamada
  European Signal Processing Conference (EUSIPCO), 2021

Invited Talks & Papers

A Systematic Overview of Speech Language Models
R. Komatsu
The Journal of the Acoustical Society of Japan, 2026

Efficient Scaling of Speech Language Models Based on Syllable-Text Interleaving
R. Komatsu
The Acoustical Society of Japan, 2026

A Comprehensive Overview of Audio Language Models
R. Komatsu
Sixth Joint Meeting Acoustical Society of America and Acoustical Society of Japan, 2025

Introduction to Multimodal Large Language Models
R. Komatsu
OTOGAKU Symposium, 2025

Work Experience

Research & Development Group, Hitachi, Ltd. Apr 2023 – Jan 2025
Applied machine learning to semiconductor manufacturing process optimization in a production setting

Patents

Search device, data compression processing device, semiconductor device manufacturing system, search method, and data compression processing method
R. Komatsu, N. Hama, Y. Mori, M. Hamamoto, M. Egi, T. Ohmori, R. Asakura, and K. Matsuda
WO Patent WO2026009272A1

Technical Skills

Languages: Python, Bash
Environments: Tsubame 4.0 supercomputer
Libraries: PyTorch, Hugging Face (transformers, datasets, trl), DeepSpeed

Awards & Scholarships

Best Student Presentation Award, Acoustical Society of Japan, 2023

Science Tokyo Tsubame Scholarship for Doctoral Students, 2025–2028